

Quantum Key Distribution

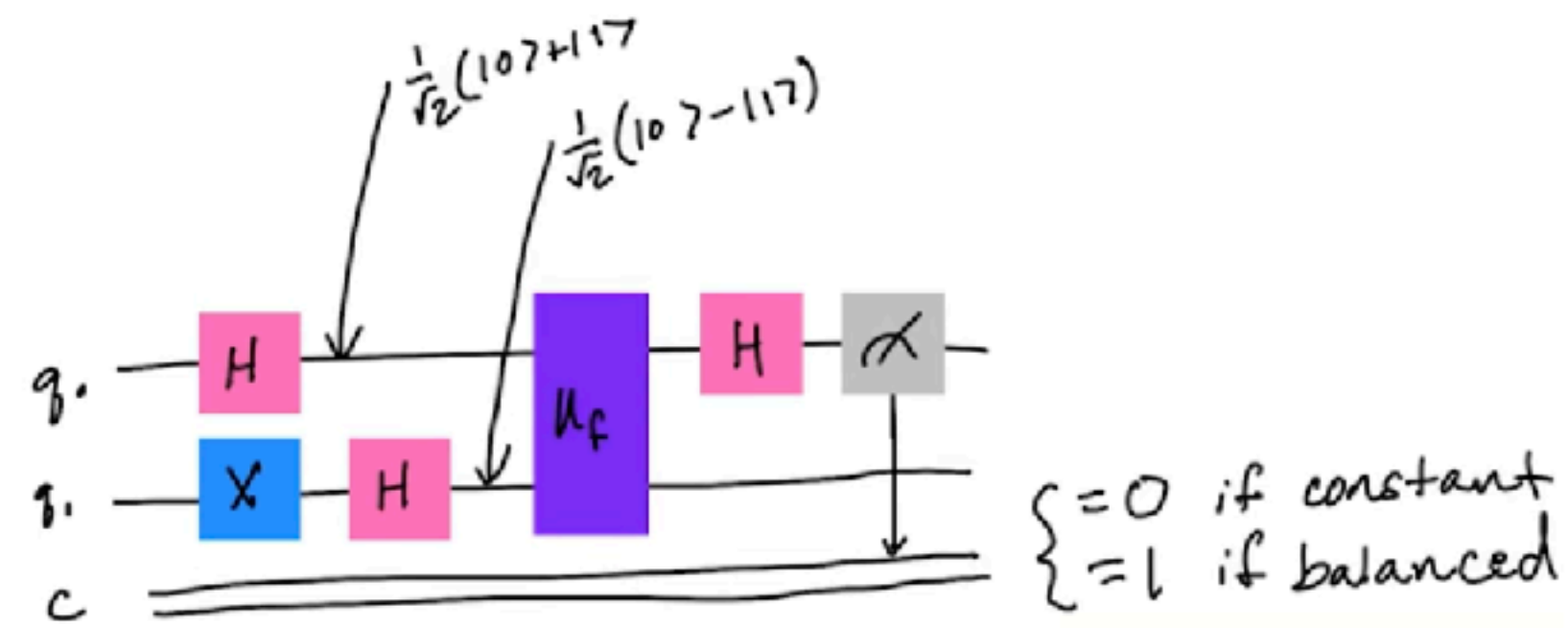
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Deutsch Algorithm

- As we know that quantum bits (qubits) can exist in superposition state of both 0s and 1s
- Query- It is the function of the input which we gives a results (in classical comp case say we feed 0, we will get either 0 or 1. Hence 2 queries including for 1 also)
- In case of quantum computer, instead of feeding a 0 or 1, it can feed a qubit that is in a superposition of 0 and 1. Hence we feeding the 0 and 1 in a single query.
- But this idea fails because once we measure the states, the superposition collapsed, we only got to measure one part of the function with a single query.
This doesn't tell us everything we need to know about which function we used.
This is called quantum parallelism which alone is not enough.
- Deutsch's algo shows us how we can use qnt plism a little differently to learn about function

x	$f_1(x)$	$f_2(x)$	$f_3(x)$	$f_4(x)$
0	0	0	1	1
1	0	1	0	1

- The bordered function columns always returns a single value regardless of the single state and these two functions are called 'constant'
- The other function will return 0 or 1 based on the input state, these functions are called 'Balanced'
- Deutsch's Algo determines if the function is constant or balanced by making some clever adjustments to the circuits we use.



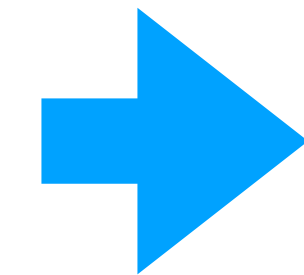
$$|\psi\rangle = \frac{1}{2}((|0\rangle-|1\rangle)|0\rangle + (|0\rangle+|1\rangle)|1\rangle)$$

$$U_f(f_2) = \text{CNOT}(q_0, q_1)$$

- Here is the circuit, for q_0 we use Hadamard and for q_1 not gate with H, preparing the qubit 1 in such a way enables the algo to use quantum paralalism in combination with interference to learn about U_f function
- In the false state, the pink is the state of qubit 0 and unordered is of state of qubit 1.
- Now both the qubits are fed in U_f ; if the function is constant, then U_f will perform some operation on q_1 alone, regardless of state of 0. Since U_f doesn't touch q_0 or care about its state, Hence the second H gate just undoes what the first H gate did.
- Look at $f_2(x)$ in the table, gives 0 when x is 0 and 1 if x is 1. On this function what U_f does to qubit 1 depends on qubit 0. If the q_0 is 0, then q_1 will be left in 0 state to get $f(x)=0$, if q_0 is 1, the U_f will apply a Not gate to q_1 and put it in a state 1.
- This is works exactly like CNOT gate which is explained in the next slide

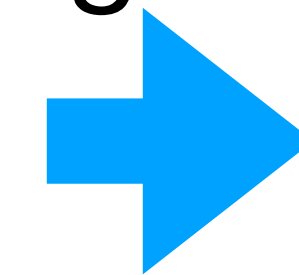
$|\Psi\rangle = \frac{1}{2}((|0\rangle - |1\rangle)|0\rangle + (|1\rangle - |0\rangle)|1\rangle)$
 $U_f(f_2) = \text{CNOT}(q_0, q_1)$

Taking
out -ve



$|\Psi\rangle = \frac{1}{2}((|0\rangle - |1\rangle)|0\rangle - (|0\rangle - |1\rangle)|1\rangle)$
 $U_f(f_2) = \text{CNOT}(q_0, q_1)$

Using a
H gate

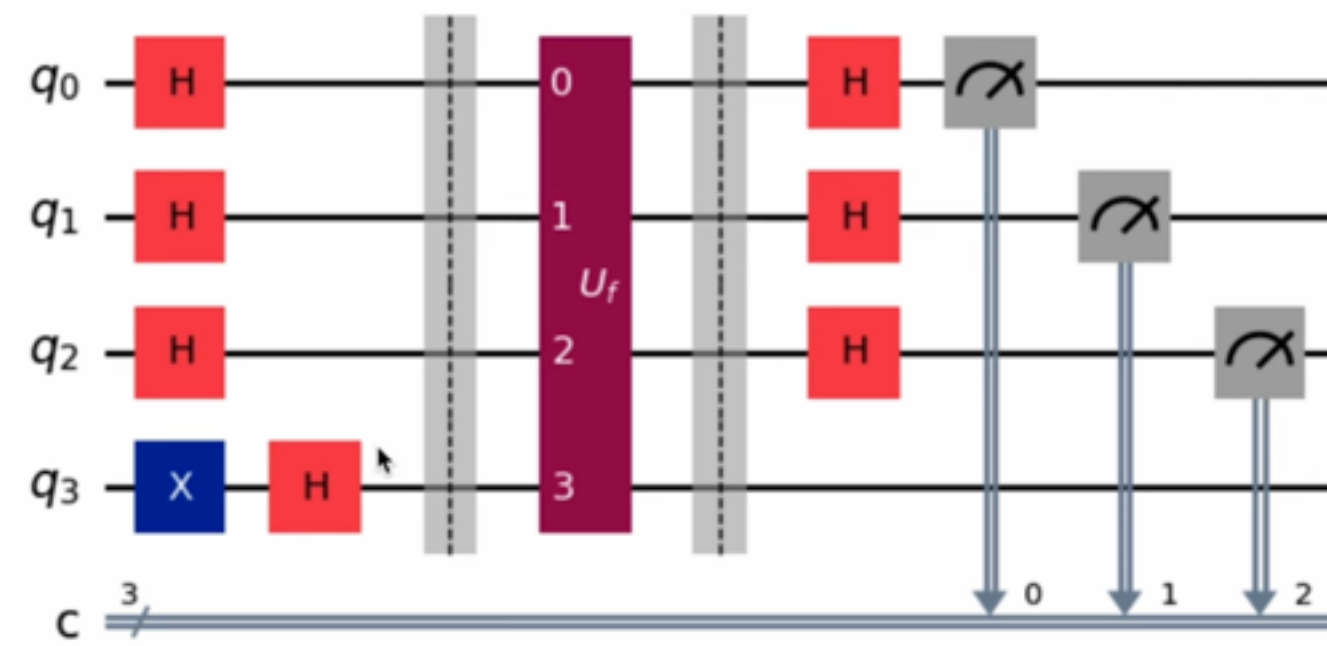


$|\Psi\rangle = \frac{1}{2}((|0\rangle - |1\rangle)|0\rangle + (|0\rangle - |1\rangle)|1\rangle)$
 $U_f(f_2) = \text{CNOT}(q_0, q_1)$

- The CNOT gate will do nothing to the first term since the control qubit is in the state 0 but on the second term (state of qubit 1), $|0\rangle$ flips to $|1\rangle$ and vice versa. Hence actually it is the state of q0 that has changed from $+|1\rangle$ to $-|1\rangle$
- When we apply the final Hadamard to q0, this transform it to state 1. Hence if the function is balanced, we will measure q0 to be in 1 state
- Hence conclusion- q0 was only used as a control qubit to change the state of qubit 1.
- Intuitively you might think that q0 would remain unchanged and yet somehow operations on qubit 1 actually change the state of qubit 0 by changing the relative phase between the two parts of its superposition. This phenomenon is known as the 'Phase kickback mechanism'.

Deutsch-Jozsa Algorithm

- Multiqubit extension of Deutsch's algorithm that demonstrates a much more impressive quantum advantage. In this case the function takes n bits and outputs a single bit but promising that the function is guaranteed to be either constant or balanced.
- The deutsch-Jozsa circuit look same like Deutsch algo but with more qubits.



- The principle remains the same, if the function is constant, it will not depend on the top 3 qubits, so their states will transform back to 0 after the second H gate, and our measurement will yield a 3-bit string of all 0s. But if the function is balanced, then when we measure the qubits following the hadamards, this phase kickback mech will cause one or more of those bits to be 1
- After we measure, we look at the measured bits to determine whether the function was balanced or constant, if they are all 0 bits, its constant and if there is even a single 1 bit, it is balanced.

Quantum Teleportation

- In quantum teleportation, information can be transmitted between two distinct locations. All you need is a pair of entangled particles, one at each site, and a way to communicate classical information between the two sites.
- Quantum information disappears from one location and appear at another distant location without travelling via intervening space.
- It is possible using the phenomenon of quantum entanglement and classical communication.
- Lets see the working in the next slide-

- Suppose we have 2 people Alice and Bob who are separated by large distance. Alice wants to send her state to Bob but she has no way to actually send the qubit itself or where to send it.
- Luckily both the parties already share a pair of entangled qubits a and b-
- By entangled, we mean that even though the two qubits are separated by large distance, their states are correlated in a way that classical physics cannot account for. This entangled pair is called an Ebit.
- Alice can use this Ebit to transmit the quantum info to Bob. It is made up of A and B qubits

$$|\psi\rangle_{BA} = \frac{1}{\sqrt{2}} (|0\rangle_B |0\rangle_A + |1\rangle_B |1\rangle_A) \quad (\text{Entangled bit})$$

Q is introduced: The secret state is prepared on qubit Q:

$$|\psi\rangle_Q = \alpha_0 |0\rangle_Q + \alpha_1 |1\rangle_Q$$

- At this point Q is simply adjacent to Alice's qubit (A). There has been no entanglement,
- Here is the joint state of all three of these qubits(A's B's and Ebit)

$$|\psi\rangle = |\psi\rangle_{BA} |\psi\rangle_Q = \frac{1}{\sqrt{2}} (|0\rangle_B |0\rangle_A + |1\rangle_B |1\rangle_A) (\alpha_0 |0\rangle_Q + \alpha_1 |1\rangle_Q)$$

The goal is to move the information on Q from Alice's location to the location of Bob.

- Now Alice will measure both of her qubits by Expanding and applying the Cnot gate followed by Hadamard gate

$$|\psi\rangle = |\psi\rangle_{BA} |\psi\rangle_Q = \frac{1}{\sqrt{2}} (|0\rangle_B |0\rangle_A + |1\rangle_B |1\rangle_A) (\alpha_0 |0\rangle_Q + \alpha_1 |1\rangle_Q)$$

$$= \frac{1}{\sqrt{2}} (\alpha_0 (|0\rangle_B |0\rangle_A + |1\rangle_B |1\rangle_A) |0\rangle_Q + \alpha_1 (|0\rangle_B |0\rangle_A + |1\rangle_B |1\rangle_A) |1\rangle_Q)$$

1. $\text{CNOT}(\text{control}, \text{target}) = \text{CNOT}(Q, A) |\psi\rangle$

$$= \frac{1}{\sqrt{2}} (\alpha_0 (|0\rangle_B |0\rangle_A + |1\rangle_B |1\rangle_A) |0\rangle_Q + \alpha_1 (|0\rangle_B |1\rangle_A + |1\rangle_B |0\rangle_A) |1\rangle_Q)$$

2 Hadamard on Q:

$$= \frac{1}{2} (\underbrace{(\alpha_0 |0\rangle_B + \alpha_1 |1\rangle_B)}_{\text{Term 1}} |0\rangle_A |0\rangle_Q + \underbrace{(\alpha_0 |0\rangle_B - \alpha_1 |1\rangle_B)}_{\text{Term 2}} |0\rangle_A |1\rangle_Q$$

$$+ \underbrace{(\alpha_1 |0\rangle_B + \alpha_0 |1\rangle_B)}_{\text{Term 3}} |1\rangle_A |0\rangle_Q + \underbrace{(-\alpha_1 |0\rangle_B + \alpha_0 |1\rangle_B)}_{\text{Term 4}} |1\rangle_A |1\rangle_Q)$$

- As a result this 3 qubit state will partially collapse into one of these 4 terms
And in each case Bob will be left with a superposition state with amplitudes (Alpha 0 and alpha 1) that are related to Alice's initial unknown state.

- Now Alice calls Bob and let know what her two measurements were. After he knows that, he will know what operations to perform on the qubits to make it match Alice's original state. Here is the possible measurements:

Alice outcome	Bob's state	Instruction to Bob	Result
$ 0\rangle_A 0\rangle_Q$	$\alpha_0 0\rangle_B + \alpha_1 1\rangle_B$	None	$\alpha_0 0\rangle_B + \alpha_1 1\rangle_B$
$ 0\rangle_A 1\rangle_Q$	$\alpha_0 0\rangle_B - \alpha_1 1\rangle_B$	Z	$\alpha_0 0\rangle_B + \alpha_1 1\rangle_B$
$ 1\rangle_A 0\rangle_Q$	$\alpha_1 0\rangle_B + \alpha_0 1\rangle_B$	X	$\alpha_0 0\rangle_B + \alpha_1 1\rangle_B$
$ 1\rangle_A 1\rangle_Q$	$-\alpha_1 0\rangle_B + \alpha_0 1\rangle_B$	X then Z	$\alpha_0 0\rangle_B + \alpha_1 1\rangle_B$

- Even though Bob's state instantly collapse Bob could get any info out of that state until Alice told Bob what she measured.

Quantum Key Distribution

- BB84 protocol, We send the key as quantum bits. Key features of which are that they obey the uncertainty principle. You can't measure quantum system without disturbing it.
- A truly random key that shifts each letter in a secret message a different random amount cannot be inferred because there are no patterns in the message that was sent.
- Only vulnerability is only that the key used doesn't fall into the wrong hands.
- Here's where qkd comes in, by using qubits, two parties can agree upon a key and be reasonably sure that a third party wasn't secretly listening.

BB84 Protocol

- The protocol starts with Alice preparing a set of qubits to be sent to Bob. These qubits become the info that will become their shared one-time pad or key.
- Alice generates two random sequences of the same length, one is a random sequence of 0s and 1s. While the other being random sequence of x's and z's representing the basis in which she will prepare her qubits.
- Then she will prepare each qubit in a quantum state according to the combination of the basis and bit value shown here.

Basis	bit = 0	bit = 1
z	$ 0\rangle$	$ 1\rangle$
x	$ +\rangle$	$ -\rangle$

- Hence, measurements on each basis only have two possible outcomes- 0 & 1 for the z basis and + or - for the x basis

- Alice now prepare a string of qubits states like-

Alice's bits	0	1	0	0	1	1	0	1	0	...
Alice's bases	X	X	Z	Z	Z	X	Z	Z	X	...
Alice's states	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	...

- Once she has prepared all of these qubits states, Alice will send Bob these qubits and Bob will randomly choose b/w x and z basis in order to measure each of these qubit states. Here is Bob measurement ex-

Bob's bases	X	Z	X	Z	X	X	Z	X	X	...
Bob's states (measured)	$ +\rangle$	$ 0\rangle$	$ -\rangle$	$ 0\rangle$	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ -\rangle$	$ +\rangle$	...
Bob's bits	0	0	1	0	0	1	0	1	0	...

- Now Bob can convert each measurement into a bit using the same table that Alice used to prepare her qubits. The second row is the measurement results of the qubits states (Photons) that Alice sent him.

- Alice encodes each classical bit (0 or 1) into a photon's quantum property, usually its Polarisation. For ex-

Bit	Basis	Polarization sent (Quantum state)
0	Z (Rectilinear)	→ (horizontal, $ 0\rangle$)
1	Z	↓ (vertical, $ 1\rangle$)
0	X (Diagonal)	↘ (45°, $ +\rangle$)
1	X	↙ (135°, $ -\rangle$)

So Alice sends a single photon whose polarisation matches her chosen bit and basis.

- Bob's goal is to measure each incoming photon and try to recover Alice's bit. Hence-
 - If Bob chooses Z basis, he uses a horizontal/vertical polarizer.
 - If Bob chooses X basis, he uses a diagonal polarizer (45° / 135°).

- Imagine Bob's detector has two possible measurement setups:

Basis	Optical Component Used	Measurement Outcomes
Z (Rectilinear)	Polarizing beam splitter aligned at 0° and 90°	Detects → or ↓ → gives bit 0 or 1
X (Diagonal)	Polarizing beam splitter rotated 45°	Detects ↘ or ↙ → gives bit 0 or 1

- Say Alice sends photon in state $|+\rangle$ (45° diagonal).
 - If Bob measures in X basis (diagonal) → detects $|+\rangle$, bit = 0 (Correct)
 - If Bob measures in Z basis (rectilinear) → photon randomly appears horizontal or vertical → random bit (wrong)

- So Bob gets the final bit string. When the basis of Alice prep and Bob's prep match, then their bits will match too.
- Way that they can both agree upon a bit string for their OTP is by sharing which bases they used for which qubit. When the basis match match, they keep the qubit.. And when they dont, they throw that away(Unconsidered).

Alice's bits	0	1	0	0	1	1	0	1	0	...
Alice's bases	X	X	Z	Z	Z	X	Z	Z	X	...
Alice's states	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	...
Bob's bases	X	Z	X	Z	X	X	Z	X	X	...
Bob's states (measured)	$ +\rangle$	$ 0\rangle$	$ -\rangle$	$ 0\rangle$	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ -\rangle$	$ +\rangle$	...
Bob's bits	0	0	1	0	0	1	0	1	0	...

- Let's see what happens when a evesdropper (eve) intercepts the qubits that Alice sent to Bob.
- Eve doesnt know which states the qubits are in or how Alice prepared them.. So because of the uncertainty principle and No-cloning theorem, Eve cant just make an identical copy of the qubits to pass along to Bob.
- Eve guesses a bases to measure each qubit, and then prepare a new set of spoof qubits to send along to Bob.
- Now, when Alice and Bob share their basis, they will decide to keep these bits for their key..

Alice's bits	0	1	0	0	1	1	0	1	0	...
Alice's bases	X	X	Z	Z	Z	X	Z	Z	X	...
Alice's states	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$	$ +\rangle$	...
Eve's guess bases	Z	X	X	Z	X	Z	Z	X	X	...
Eve's states (measured)	$ 1\rangle$	$ -\rangle$	$ +\rangle$	$ 0\rangle$	$ -\rangle$	$ 0\rangle$	$ 0\rangle$	$ -\rangle$	$ +\rangle$	...
Eve's states (sent on)	$ 1\rangle$	$ -\rangle$	$ +\rangle$	$ 0\rangle$	$ -\rangle$	$ 0\rangle$	$ 0\rangle$	$ -\rangle$	$ +\rangle$	...
Bob's bases	X	Z	X	Z	X	X	Z	X	X	...
Bob's states (measured)	$ -\rangle$	$ 0\rangle$	$ +\rangle$	$ 0\rangle$	$ -\rangle$	$ +\rangle$	$ 0\rangle$	$ -\rangle$	$ +\rangle$	...
Bob's bits	1	0	0	0	1	0	0	1	0	...

but the bits dont match as Bob measured the qubits that Eve prepared not Alice.. Hence they can assume eve